MULTIMODAL AI TUTORING SYSTEM

25-26J-130

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science

Sri Lanka Institute of Information Technology
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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

This study aims to create and implement a privacy-focused, real-time engagement monitoring system that uses Computer Vision (CV) and Vision-Language Models (VLM) within a multimodal intelligent tutoring system. It addresses the shortcomings of current engagement detection methods, which depend only on visual analysis, lack an understanding of the learning context, and often raise privacy concerns due to cloud-based processing. The proposed system uses lightweight CV models to capture behavioral signals such as facial expressions, gaze, and posture from a learner's webcam in real time. At the same time, a VLM interprets the meaning of on-screen learning material obtained through OCR or image capture. These outputs combine to form a single "engagement-context state" that connects emotional and cognitive indicators to specific lesson content. This information is then processed by a Reinforcement Learning (RL) engine and an Agentic AI core, enabling proactive and adaptable teaching strategies. By integrating behavioral and semantic insights within a privacy-aware, on-device framework, this research aims to connect affective computing and teaching adaptation. It offers a new, ethical, and effective solution for the next generation of intelligent tutoring systems.

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1. Introduction

The rapid growth of artificial intelligence (AI) has changed educational technology, allowing the creation of Multimodal AI Tutoring Systems that respond to each learner's cognitive, emotional, and behavioral states. This system uses various components, such as natural language processing, computer vision, reinforcement learning, and agentic planning, to provide personalized, context-aware, and emotionally intelligent teaching. By using different data sources and smart decision-making frameworks, multimodal tutors aim to mimic the flexibility and responsiveness of human teachers in digital settings.

The core component of this research is the Computer Vision Monitoring system combined with Vision-Language Models (VLMs). This module assesses student engagement in real time by analyzing visual signals like facial expressions, gaze, and posture, and connecting them to the learning context through semantic reasoning. Unlike traditional systems that depend only on visual classification, in the educational domain, this combined method can identify subtle engagement levels, such as "confused by fractions" or "bored during reading," and offers meaningful feedback to the tutoring engine. This integration improves the tutor's ability to adjust instruction dynamically, making learning more personalized, effective, and ethically responsible.

To effectively implement this system, several key techniques and studies must be learned. In the area of computer vision, foundational methods like facial expression recognition, gaze tracking, and posture analysis are crucial, with models such as YOLOv9 [3] and frameworks like MediaPipe enabling efficient real-time detection. Temporal modeling techniques, including Vision Transformers (ViTs) and Temporal Convolutional Networks (TCNs) [2], are essential for capturing engagement patterns over time. On the VLM side, it is important to master models like CLIP [5], LLaVA [6], and BLIP [7], as they help interpret visual signals by connecting them to textual content, bridging the gap between behavioral indicators and instructional relevance. Furthermore, understanding multimodal fusion strategies and privacy-preserving AI practices is vital for combining these elements into a cohesive, ethical, and effective tutoring system.

1.1. Background & Literature Survey

Computer Vision in Engagement Detection

Computer Vision (CV) has become essential in educational research for monitoring student engagement. It helps systems interpret non-verbal signals like facial expressions, gaze direction, and posture. These indicators reveal attention, confusion, and emotional state. Such visual signals are especially useful in digital learning settings where traditional teacher-student interactions are limited.

Introduced a bagging ensemble deep learning method for detecting student engagement in online learning[1]. Their model showed better accuracy and reliability across different datasets by analyzing facial and behavioral cues. presented a hybrid model that combined Vision Transformers (ViTs) with Temporal Convolutional Networks (TCNs) to capture both spatial and temporal features of student behavior [2]. This approach improved the detection of engagement patterns over time, particularly in asynchronous learning environments.

Used YOLOv9, a real-time object detection model, to monitor attention in classroom settings [3]. Their system accurately detected gaze direction and posture, allowing for immediate feedback on student attention levels. The lightweight CV models made the system suitable for real-world educational settings, where efficiency and quick responses are crucial.

Performed a thorough review of machine vision methods for engagement monitoring [4]. They discussed the benefits of deep learning approaches while pointing out key drawbacks, such as the lack of semantic reasoning, fragmented system designs, and privacy issues. These drawbacks highlight the need for systems that extend beyond visual classification and incorporate contextual understanding and ethical practices.

Vision-Language Models in Education

VLMs mark an important step forward in multimodal AI. They allow systems to connect visual inputs with text descriptions. Models like CLIP [5], LLaVA [6], and BLIP [7] can interpret images semantically, making them very useful for educational purposes.

CLIP learns to understand visual information through natural language supervision[5]. This lets it link images with text concepts. LLaVA, builds on this by modifying vision-language models for task-based instructions, making it fit for generating educational feedback [6]. BLIP, emphasizes a unified approach to understanding and generating vision-language content[7]. This is crucial for providing adaptive and context-aware responses in tutoring.

Even though these models have worked well for general image-text reasoning tasks, their application in real-time educational settings is still being explored. These models can help connect behavioral detection with teaching relevance. For example, a system could notice if a student looks confused and, using VLMs, deduce that the confusion relates to a particular math concept being taught. This connection allows for more focused and adaptive feedback, improving how a tutor can respond to student needs.

Combining VLMs with computer vision creates multimodal fusion. This process merges behavioral signals, like facial expressions and gaze, with contextual information, such as the current topic or question. This fusion helps produce detailed engagement states that are grounded both emotionally and cognitively. Additionally, using on-device processing and anonymized feature extraction keeps the system compliant with privacy standards, making it appropriate for use in real classrooms.

1.2. Research Gap

Despite progress in intelligent tutoring systems, several significant gaps remain in detecting and managing student engagement.

Visual-Only Engagement Detection: Current methods mainly rely on computer vision cues such as gaze direction, posture, and facial expressions. While these can identify if a student seems engaged or disengaged, they do not explain why disengagement happens or how it connects to the learning content [1-4].

Underutilization of Vision-Language Models (VLMs): Models like CLIP, LLaVA, and BLIP can effectively connect visual and textual data. However, their use in real-time educational settings is still limited. In particular, we have not yet taken full advantage of how they can link student behaviors, such as frowning, to learning content, like algebra problems [5–7]

Fragmented Architectures: Current engagement monitoring tools often work independently from tutoring logic, providing reactive feedback instead of integrated and adaptive teaching strategies. This disconnection limits how personalized learning experiences can be in real-time [1-4].

Privacy and Ethical Concerns: Many current systems rely on cloud-based video processing. This raises ethical and legal issues in both classroom and home settings. There is a lack of privacy-aware, on-device solutions that protect sensitive student data [8].

Limited Adaptivity in Current AI Tutors: Commercial AI tutoring systems like vTutor, AutoTutor, TutorOcean AI, and Cognispark AI show the potential for adaptive learning but are mostly text-focused and reactive. They do not include live engagement monitoring or multimodal feedback, which limits their ability to adapt instruction dynamically [9-11].

These gaps emphasize the need for a real-time, multimodal, and privacy-aware engagement monitoring system. Such a system should integrate computer vision and vision-language models, combine behavioral and content-based insights, and feed them into an adaptive Agentic AI Core with reinforcement learning for meaningful, ethical, and dynamic tutoring

1.3. Research Problem

The main research question in this proposal is:

How to create a real-time, multimodal, privacy-focused engagement monitoring system that connects student behaviors to learning content and provides actionable feedback to adaptive tutoring engines?

This question includes several sub-problems:

Linking Behaviors to Context: How can we better interpret student behaviors like gaze, posture, and facial expressions by linking them directly to the online learning context instead of viewing them as simple engagement signals?

Adapting Vision-Language Models (VLMs): How can we modify VLMs for lightweight, real-time use in virtual classrooms to offer insights into student engagement without needing a lot of computing power?

Multimodal Integration into Adaptive Loops: How can we combine CV-based and VLM-based engagement monitoring into the adaptive loop of an intelligent tutoring system, allowing for proactive changes to teaching strategies rather than just responding to disengagement after it happens?

Ensuring Privacy and Ethics: How can we design engagement monitoring to prioritize privacy through on-device or anonymized processing while still providing reliable and useful insights in online classroom environments?

Actionable Feedback Delivery: How can we deliver engagement insights to the tutoring system clearly and relevantly, ensuring that they support the tutor's decision-making without overwhelming it with raw data?

Addressing this research problem will help create next-generation intelligent tutoring systems that move beyond basic monitoring. The proposed system will provide context-aware, multimodal, adaptive, and privacy-protecting feedback, improving personalization, engagement, and learning outcomes in online classrooms.

2. Objectives

2.1 Main Objective

The main goal of this research project is to develop a multimodal intelligent tutoring system. This system will combine Agentic AI, Reinforcement Learning (RL), Large Language Models (LLMs), and Computer Vision (CV) with a Vision Language Model (VLM) to improve online education. Unlike current systems that mainly react to input or operate in a limited way, the proposed system seeks to provide personalized, proactive, and adaptable learning experiences. It will understand the learner's cognitive and emotional states, adjust teaching strategies in real-time, and create content that fits the learner's situation and needs. By merging these four AI technologies into one framework, the system aims to imitate the function of a skilled human tutor. This includes long-term planning, real-time adjustments, and empathetic responses, all while ensuring it can be scaled and deployed ethically in online learning settings.

2.2 Sub Objective

These specific objectives describe the main parts of the system. Each objective supports the overall goal. Together, they make sure that the tutor works as an integrated, smart, and learner-focused platform.

1. Develop an Agentic AI Core for adaptive planning

- Create a memory-based, goal-driven tutoring logic that can plan teaching strategies over extended periods, similar to a proactive human tutor.
- Use methods like Tree of Thought or ReAct-based reasoning so the tutor can make multi-step decisions. This will help learners follow clear and effective learning pathways.
- Test this component to ensure it can maintain planning coherence, adjust goals as needed, and improve task completion efficiency in complex online learning scenarios.

2. Design a Reinforcement Learning Engine for personalized adaptation

- Build a reinforcement learning framework that adjusts teaching strategies based on real-time engagement, performance, and retention metrics.
- Define a custom reward system that balances cognitive gains with student motivation. This ensures that interventions improve both short-term engagement and long-term learning outcomes.
- Train the RL model using simulated learner profiles and real feedback from actual students. Validate its performance by comparing it to traditional tutoring methods that rely on heuristics.

3. Develop an LLM-Driven Content Generator for dynamic instruction

- Fine-tune a Large Language Model to generate customized content such as quizzes, hints, summaries, and explanations based on the student's current knowledge, curriculum requirements, and emotional state.
- Introduce prompt tuning and control methods to adjust the generated material's tone, style, and difficulty level according to the learner's profile and mood.
- Check the educational quality of the content by conducting controlled studies and measuring improvements in learning retention, comprehension, and student satisfaction compared to static learning materials.

4. Implement a Computer Vision Attention Monitor integrated with Vision-Language Models.

- Build a lightweight, privacy-aware computer vision system that can detect gaze direction, posture, and facial expressions to determine engagement levels in real time.
- Extend this by adding Vision-Language Models (e.g., CLIP, LLaVA) so that visual cues relate to the learning material. This helps the tutor identify states like "confused by algebraic equations" or "bored during history lesson" instead of just marking disengagement.
- Ensure privacy and ethical use by using on-device inference and anonymized feature extraction. This prevents sensitive video data from being stored or sent elsewhere.
- Validate this component by looking at the links between visual engagement signals, contextual interpretations, and measurable improvements in learning outcomes.

3. Methodology

The suggested system can be divided into four main parts:

Agentic AI Core-:This includes goal-oriented tutor logic and memory-based planning.

Reinforcement Learning Engine-: is an adaptive policy learner that focuses on engagement and performance.

LLM Content Generator-: This generates dynamic personalized learning materials.

Computer Vision and VLM Engagement Monitor-: This detects and interprets real-time multimodal engagement.

This involves collecting relevant datasets, preparing the data, implementing the necessary models, training them, arranging the parts logically, and integrating them to work together as a unified system.

3.2 Overall System

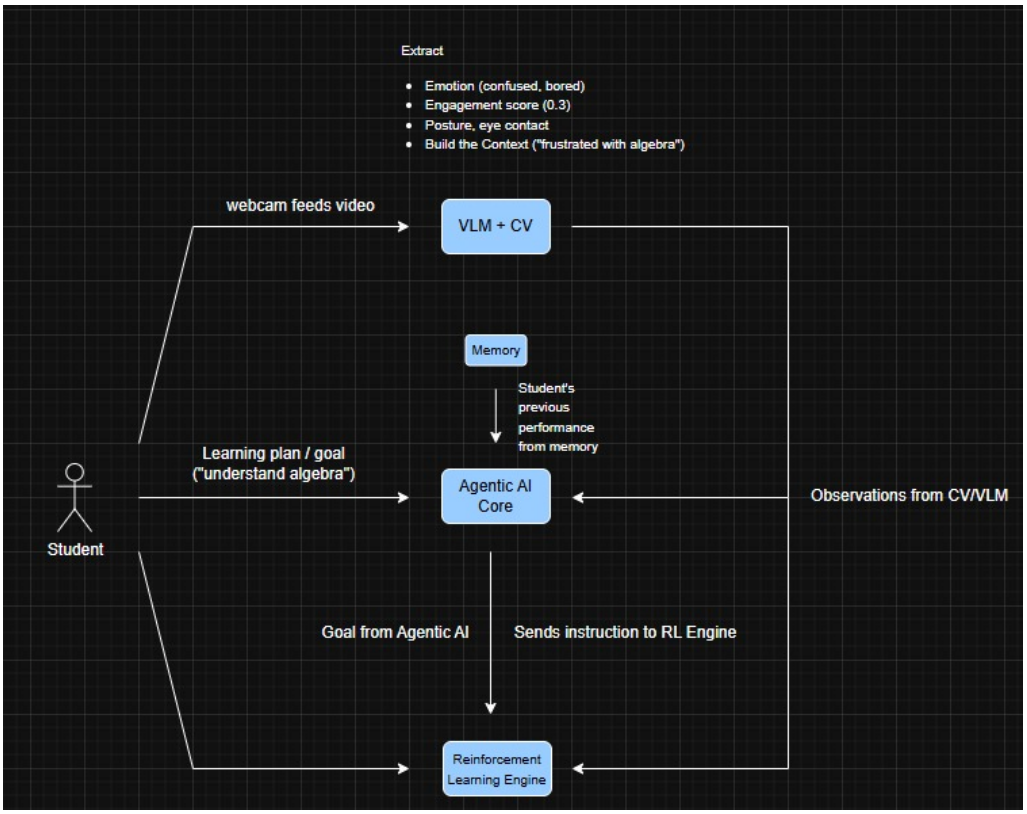

Figure 1: Overall System Diagram

The "Multimodal AI Tutoring System Architecture" diagram outlines the structure of a system that provides adaptive, intelligent, and personalized online tutoring sessions with real-time monitoring and feedback. Here's a detailed look at each component:

Learner Interaction

Web Application: Learners use a web application to access the system. This platform is the main hub where learners can access lessons, complete exercises, receive real-time explanations, and view progress updates.

Tutor Interaction (AI Core)

The tutor logic, driven by the Agentic AI Core, acts as the central brain of the system. It tracks learner goals, progress history, and session context. This ensures the teaching strategies are personalized and focused on goals.

Initial Diagnosis and Adaptive Planning

Before each learning session, the system assesses the learner's knowledge and engagement. The Agentic AI Core, with input from the Reinforcement Learning Engine, uses this information to create personalized learning paths and adjust lesson schedules

Video and Activity Monitoring

The system connects to the learner through a real-time interface, allowing continuous monitoring of engagement. This live interaction provides important data for the Computer Vision (CV) and Vision-Language Models (VLMs), which analyze gaze, posture, and expressions alongside the learning material.

Computer Vision and Engagement Analysis

During the session, the CV model (YOLOv9, MediaPipe) identifies behavioral signals, including facial expressions, gaze, and posture. These signals help determine attention levels and emotional states, providing insight into how learners engage with the content.

Vision-Language Model Integration

At the same time, the VLMs (CLIP, LLaVA) link the visual signals with the learning material. For instance, if a student appears confused, the VLM assesses whether the confusion relates to fractions, algebra, or another concept. This ensures a deep understanding of engagement.

Content Generation via LLM

The LLM-driven Content Generator creates personalized hints, examples, quizzes, and explanations. It adjusts difficulty, tone, and style based on the learner's progress, emotional state, and the teaching strategies set by the Agentic AI Core.

Reinforcement Learning Engine

The RL Engine continually improves instructional strategies using real-time feedback on engagement and performance. It updates the teaching approach on the fly, ensuring that learners stay engaged, motivated, and make steady progress.

Integrated Feedback Loop

All detected signals and insights are sent back to the Agentic AI Core and RL Engine. Updated strategies are applied, and personalized content is generated for the learner. This loop continues dynamically, creating a proactive, adaptive, and privacy-conscious tutoring experience.

Overall Impact

This architecture creates a detailed, multimodal, and privacy-aware tutoring system that not only responds to learner performance but also anticipates needs, delivers meaningful feedback, and ensures tailored instruction in online classrooms.

3.2 Individual System Diagram

The project seeks to develop a system that recognizes student engagement levels in online learning by analyzing visual and contextual input in real time. The system will use a combination of Computer Vision (CV) models, including MediaPipe, along with Vision-Language Models (VLMs) like CLIP and LLaVA. It will detect and interpret behavioral signals such as facial expressions, gaze, and posture, and connect them to the learning content. The process captures visual data and generates insights so that the tutoring engine receives continuous feedback. This feedback allows it to change learning strategies as needed.

System Overview and Diagram

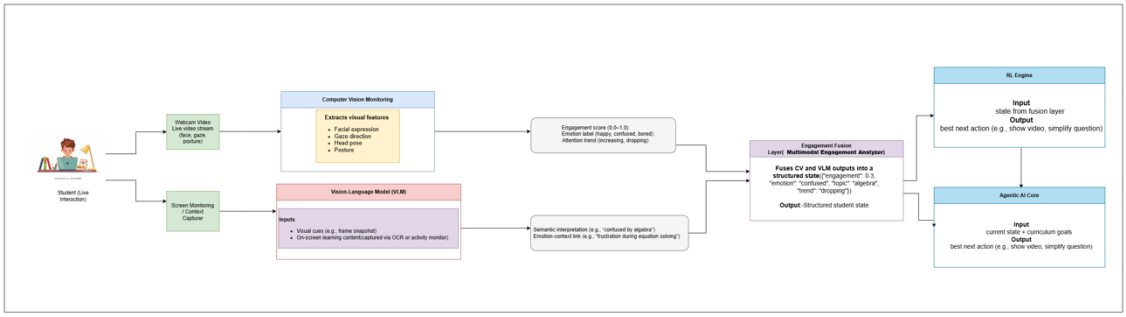

Figure 2: Individual System Diagram

The proposed system consists of both front-end and back-end components, as shown in the diagram. The student can join the session using either a mobile or desktop device, while the tutoring system processes real-time visual and contextual data. The tutor interface displays analyzed results, such as “student confused during fractions,” which enables adaptive teaching.

Video Input: The student's webcam stream is captured through the online tutoring platform, and the raw video feed is sent to the back-end system for analysis.

Pre-Processing: The video feed goes through pre-processing steps that include frame extraction, face detection, normalization, and anonymization for privacy. Lightweight CV libraries like MediaPipe are used to extract key points such as gaze, head pose, and micro-expressions.

Feature Extraction: Behavioral features are extracted from the processed frames. These include gaze direction, blink rate, facial expressions, posture changes, and attention spans. These features serve as indicators of focus and emotional state.

Contextual Feature Alignment (VLMs): The system pairs the extracted behavioral signals with the learning content displayed on the screen using CLIP or LLaVA. This ensures that engagement cues are connected to the actual material. For example, if a student appears confused while viewing an algebra problem, the system identifies that the confusion relates to the content.

Model Selection and Training: A multimodal fusion model will be trained to merge CV signals with contextual embeddings from VLMs. This hybrid approach enables the system to provide specific insights like “student is disengaged during geometry” instead of general statements like “student not attentive.”

Real-Time Data Service: The combined output is sent to the Reinforcement Learning Engine and Agentic AI Core in real time. This allows the tutoring system to change strategies immediately, such as switching examples, providing hints, or slowing down explanations.

Integration: The CV and VLM module is linked with other system components, such as the Agentic AI, RL engine, and LLM content generator. This setup creates a feedback loop where student behavior continually informs teaching strategies.

Implementation Approach

The project will be conducted in several phases. It will focus on collecting, processing, and integrating real-time webcam data into the tutoring system.

Phase 1: Webcam Data Capture and Pre-Processing

Set up the video streaming pipeline to capture real-time student webcam input.

Apply pre-processing steps like frame extraction, face detection, and noise reduction to ensure clean input for analysis.

Phase 2: Computer Vision (CV) Analysis

Use lightweight CV models, such as MediaPipe, to extract behavioral cues like gaze, posture, and facial expressions.

Generate indicators of attention and engagement from the extracted features.

Phase 3: Vision-Language Model (VLM) Integration

Feed contextual learning content, such as the displayed question and topic, along with CV outputs into VLMs like CLIP and LLaVA.

Interpret visual engagement cues in relation to the specific learning material to derive insights, such as a student being confused by an algebra equation.

Phase 4: Multimodal Fusion and Hybrid Model Development

Combine CV-based behavioral signals with VLM contextual insights into a single multimodal representation.

Ensure the fusion model operates in real time with minimal delay.

Phase 5: Adaptive Feedback via RL Engine and Agentic AI Core

Pass the multimodal insights into the Reinforcement Learning (RL) engine.

The RL engine updates strategies to improve teaching methods based on engagement feedback.

The Agentic AI Core uses planning to dynamically adjust lesson flow, provide hints, or change difficulty:

Tasks and Sub-Tasks

1. **Video Capture and Pre-Processing:**
    - Integrate webcam input with the tutoring platform.
    - Integrate webcam input with the tutoring platform..
2. **Feature Extraction:**
    - Extract behavioral cues like gaze, emotion, and posture.
    - Encode the context of learning materials.
3. **Model Development and Training:**
    - Train a CV and VLM multimodal fusion model.
    - Fine-tune it for real-time educational tasks.
4. **Real-Time Analysis:**
    - Integrate live engagement predictions into the RL engine.
5. **System Integration and Testing:**
    - Connect the CV and VLM module with the AI Core and LLM.
    - Test it with real online learning sessions.

Materials and Tools Required

Hardware: A desktop or laptop with a webcam for learners. A GPU-enabled server for training.

Software: Python, PyTorch, MediaPipe, CLIP, LLaVA, and Flask API for integration.

Data Collection and Analysis

The system relies entirely on real-time video input captured from the student's webcam during online learning sessions. It does not use any pre-recorded or third-party datasets. The captured video stream serves as the main data source for the Computer Vision (CV) and Vision-Language Model (VLM) modules.

Data Capture: The student's webcam feed records continuously during the session. Frames are extracted in real time for processing.

Feature Processing: The CV models (YOLOv9, MediaPipe) analyze facial expressions, gaze direction, and posture to identify signs of attention or disengagement.

Contextual Mapping: The VLM models (e.g., CLIP, LLaVA) interpret these behavioral cues based on the specific learning content displayed on the student's screen at that time.

Analysis Pipeline: The combined CV and VLM outputs provide insights at a semantic level (e.g., "student confused during algebra problem"). These insights are immediately sent to the reinforcement learning engine and agentic AI core for an adaptive response.

Ethical Handling: Since the system works with real-time streams, data is processed on-device or in secure session pipelines without saving raw video. Only anonymized features and engagement metrics are kept for session reports.

This approach ensures that engagement detection and feedback are context-aware, real-time, and consider privacy, while eliminating reliance on pre-labeled datasets.

4. Project Requirements

4.1 Functional Requirements

Functional requirements define the main functions that the system must support to achieve its goals. For this project, the primary functional requirements are,

Real-Time Webcam Data Capture:

The system must continuously capture live video from the student's webcam during online sessions. This video is the main source for analyzing both behavioral engagement cues.

Computer Vision Based Engagement Recognition:

The system must analyze the student's visual cues, such as gaze direction, facial expressions, and posture, using CV models. These models will detect signs of attention, distraction, confusion, or disengagement in real time.

Vision Language Model (VLM)-Based Content Understanding:

The application must process and interpret details from the learning material using VLMs. This allows the system to connect observed disengagement or confusion with specific concepts, like algebra equations or reading comprehension tasks.

Multimodal Fusion:

The system must combine the outputs of the CV models and VLM models into a multimodal hybrid framework. This fusion will allow for a more complete analysis of the student's engagement by merging behavioral signals with content-specific understanding.

Adaptive Feedback Delivery via RL and Agentic AI Core:

The system must input the multimodal outputs into the Reinforcement Learning engine and Agentic AI Core. Based on this information, the tutor should adjust the teaching strategy, provide personalized hints, or change lesson difficulty in real time.

Privacy-Aware Data Processing:

All captured webcam data must be processed with strong privacy protections. This includes local or on-device processing or anonymization to ensure compliance with ethical standards and data protection laws.

4.2 Non-Functional Requirements

Non-functional requirements define the system's attributes, focusing on how well it performs under specific conditions rather than what it does

Accuracy

The system must achieve a high level of accuracy in detecting student engagement and interpreting contextual learning cues. Both the CV and VLM models should be trained and validated with live data to ensure reliable performance across different student demographics, lighting conditions, and learning materials.

Real-Time Processing

The system must process webcam input and generate multimodal engagement insights with minimal delay. This is critical to ensure that the RL and Agentic AI Core can change teaching strategies immediately during live online classes.

Usability

The user interface should be simple and clear so that teachers and students can interact with the system without needing advanced technical skills. Visualizations of engagement levels and adaptive feedback should be easy to understand.

Security

Strong security measures must be in place to protect student data. This includes end-to-end encryption for webcam streams, anonymization of sensitive data, and secure access control to prevent unauthorized access to student information.

Scalability

The system should support multiple online classes at the same time and analyze engagement data from several students without losing performance. It should scale efficiently in both cloud and on-premises environments.

Reliability

The system must operate with at least 95% uptime during live sessions to ensure continuity in the learning experience. Automatic fallback mechanisms should be ready in case of minor failures to avoid interrupting classes.

Compliance

The system must meet relevant educational data protection and privacy regulations (e.g., GDPR, COPPA, FERPA). This ensures that student video data and interaction logs are handled legally and ethically.

Extensibility

The architecture should be flexible enough to include future AI components, such as vocal emotion recognition or biometric signals, allowing the system to evolve into a more complete multimodal tutor.

4.3 User Requirements

Student Requirements

For students, the system should be easy to use, lightweight, and respect privacy. Engagement tracking through webcam and interaction analysis should not disrupt learning. Students should feel secure knowing their video and learning data are safely stored, anonymized when possible, and used only for education. The system should also offer personalized learning paths and adaptive feedback so that each student gets support suited to their pace and learning style.

Accessibility

The application should work on multiple devices and platforms, including desktops, laptops, and mobile devices, to fit different classroom setups. The system should support common online learning environments, like Zoom, MS Teams, Google Meet, and LMS integrations such as Moodle and Blackboard, so students and teachers can connect without extra technical issues. Additionally, the interface should follow universal design principles to assist users with disabilities. This includes text-to-speech compatibility, clear fonts, and customizable display options.

4.4 System Requirements

- **Hardware**

- Student side: Laptop/PC with webcam (minimum 720p).
- Server side: GPU-enabled server for running CV and VLM models.

- **Software**

- **FrontEnd:** React.js (for web app), Tailwind (UI).
- **BackEnd:** Spring Boot (API), Flask (ML model serving).
- **Models:** YOLOv9, MediaPipe, CLIP, LLaVA.

- **Network**

- Minimum 5 Mbps stable internet for video transmission.

Gantt Chart

	Task	Start Date	Due Date
1	Research Project	2025/03/24	2026/06/30
2	Brainstorming	2025/03/25	2025/03/26
3	Supervisor selection and group registration	2025/05/26	2025/05/27
4	TAF Submission	2025/06/30	2025/07/16
5	Proposal Presentation	2025/07/08	2025/07/12
6	Proposal Submisson	2025/07/19	2025/07/20
7	50% Completion	2025/12/15	2026/12/19
8	PP1	2025/12/15	2026/12/19
9	80% Completion	2026/03/01	2026/06/30
10	PP2	2025/03/01	2026/06/30
11	Final Presentation	2025/05/01	2026/06/30
12	Final report submisson	2025/06/01	2026/06/30

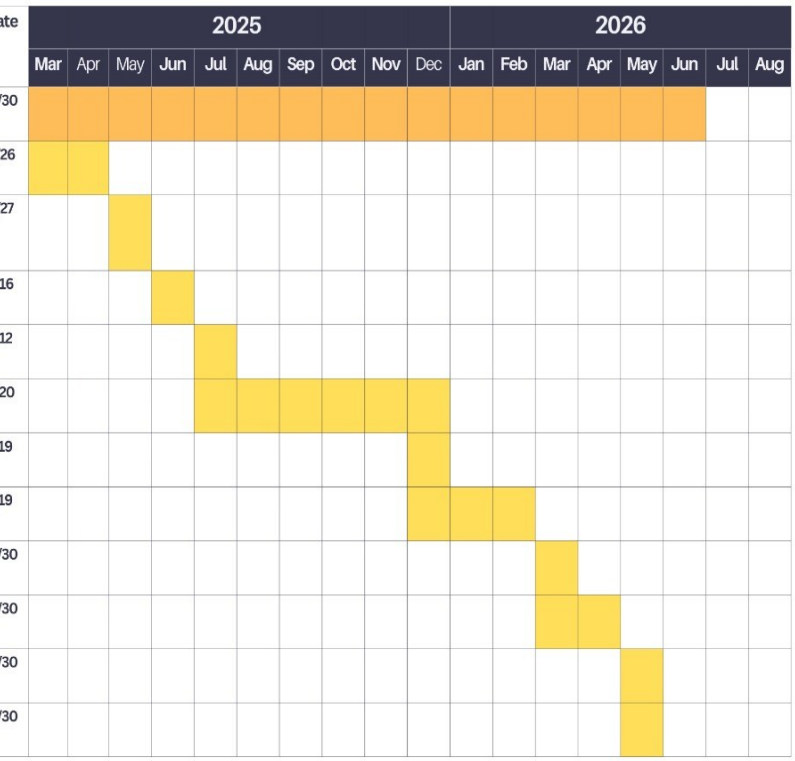

Figure 3:Gantt Chart

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